

Chapter 8 – Work and Kinetic Energy

UM-SJTU Joint Institute
Physics I (Summer 2024)
Mateusz Krzyzosiak

1 Motivation

2 Work

- How to Calculate Work in Various Situations?
- Example: Work Done on/by a Spring
- Examples: Work Along a Curved Path

3 Work and Motion

- Kinetic Energy and Work-Kinetic Energy Theorem
- Power

Motivation

Mechanics

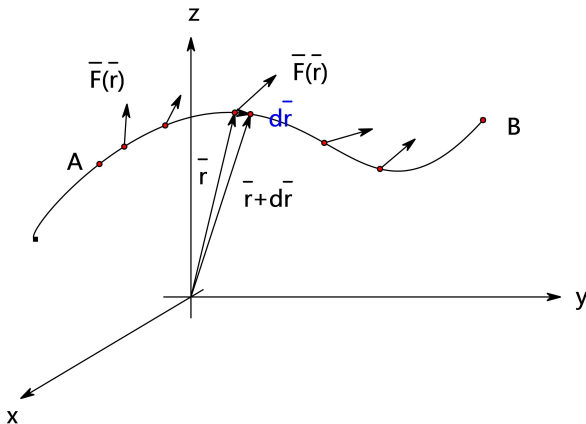
- Kinematics ($\vec{r}, \vec{v}, \vec{a}$) *HOW?*
- Dynamics ($\vec{F}, \vec{a} = \frac{\vec{F}}{m}$) *WHY?*

New scalar quantities

- Concept of *work*: How to measure the "effort" put into moving the particle from one place to another?
- *Kinetic energy* and *work-kinetic energy theorem*: How does this "effort" change the state of the particle (e.g. its speed)?

Work

Work



In general,

$$\vec{F} = \vec{F}(\vec{r})$$

(position-
dependent
force;
vector field)

Elementary work done by $\vec{F}$ when the particle moves from $\vec{r}$ to $\vec{r} + d\vec{r}$

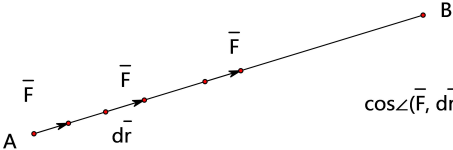
$$\delta W = \vec{F} \circ d\vec{r}$$

Total work done by $\vec{F}$ when the particle moves from A to B along the curve Γ_{AB} — add all infinitesimal contributions

$$W_{AB} = \int_{\Gamma_{AB}} \vec{F} \circ d\vec{r}$$

How to calculate work in various situations?

(A) Constant force in the direction of the displacement;
displacement along a straight line

$$\delta W = \vec{F} \circ d\vec{r} = |\vec{F}| \cdot |d\vec{r}|$$


$\cos \angle(\vec{F}, d\vec{r}) = 1$

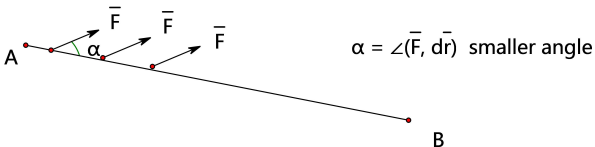
$$W_{AB} = \int_{\Gamma_{AB}} \vec{F} \circ d\vec{r} = \int_{\Gamma_{AB}} |\vec{F}| |d\vec{r}| = |\vec{F}| \underbrace{\int_{\Gamma_{AB}} |d\vec{r}|}_{S_{AB}} = |\vec{F}| S_{AB},$$

where S_{AB} is the length of AB . Hence, in this case

$$W_{AB} = |\vec{F}| S_{AB}$$

(B) Constant force acting at an angle to the direction of straight-line displacement

$$\begin{aligned}\delta W &= \vec{F} \circ d\vec{r} \\ &= |\vec{F}| \cdot |d\vec{r}| \cos \alpha\end{aligned}$$



$$\begin{aligned}W_{AB} &= \int_{\Gamma_{AB}} \vec{F} \circ d\vec{r} = \int_{\Gamma_{AB}} |\vec{F}| \cdot |d\vec{r}| \cos \alpha = |\vec{F}| \cos \alpha \underbrace{\int_{\Gamma_{AB}} |d\vec{r}|}_{S_{AB}} \\ &= |\vec{F}| S_{AB} \cos \alpha\end{aligned}$$

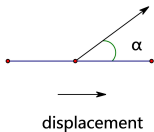
Hence, in this case,

$$W_{AB} = |\vec{F}| S_{AB} \cos \alpha$$

Conclusion: Work may either positive or negative or zero.

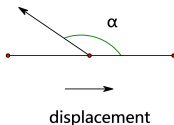
Illustration. Positive/negative work.

(a) $W_{AB} > 0$



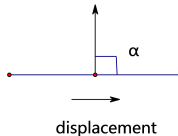
$$0 \leq \alpha < \frac{\pi}{2}$$

(b) $W_{AB} < 0$



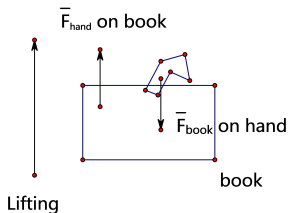
$$\frac{\pi}{2} < \alpha < \pi$$

(c) $W_{AB} = 0$



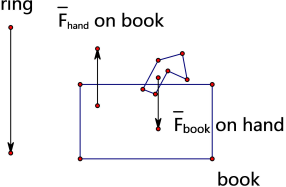
$$\alpha = \frac{\pi}{2}$$

Remember what does work on what!



$$\begin{aligned} W_{\text{book on hand}} &< 0 \\ W_{\text{hand on book}} &> 0 \end{aligned}$$

Lowering



$$\begin{aligned} W_{\text{book on hand}} &> 0 \\ W_{\text{hand on book}} &< 0 \end{aligned}$$

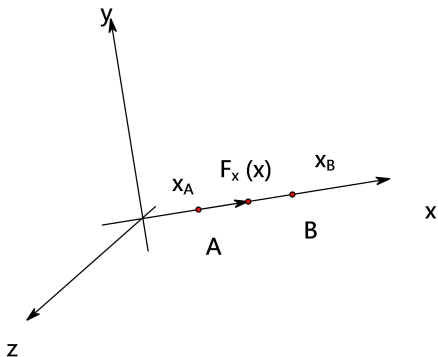
Remark. Elementary work done in the case when multiple forces act on a particle

$$\begin{aligned}\delta W &= (\bar{F}_1 + \bar{F}_2 + \dots + \bar{F}_N) \circ d\vec{r} = \bar{F}_1 \circ d\vec{r} + \bar{F}_2 \circ d\vec{r} + \dots + \bar{F}_N \circ d\vec{r} \\ &= \delta W_1 + \delta W_2 + \dots + \delta W_N\end{aligned}$$

or, equivalently,

$$\delta W = \bar{F}_{\text{net}} \circ d\vec{r}$$

(C) Force of varying magnitude acting along a straight-line path



$$\vec{F} = (\underbrace{F_x(x)}_{\text{function of } x \text{ only}}, 0, 0)$$

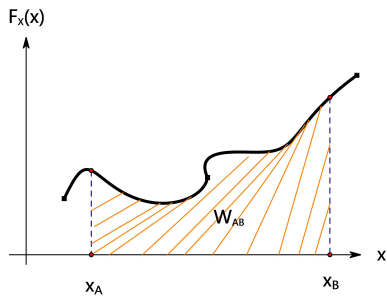
and

$$d\vec{r} = (dx, 0, 0)$$

$$\delta W = \vec{F} \circ d\vec{r} = F_x(x) dx \quad (\text{takes care of the sign})$$

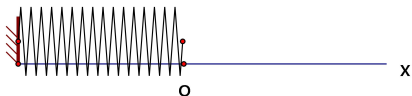
$$W_{AB} = \int_{\Gamma_{AB}} \delta W = \int_{\Gamma_{AB}} F_x(x) dx = \int_{x_A}^{x_B} F_x(x) dx$$

Interpretation (area under the curve)

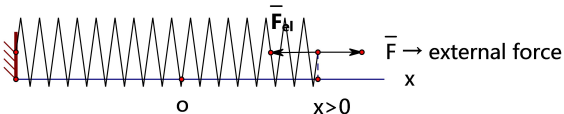


Example: Work Done on/by a Spring

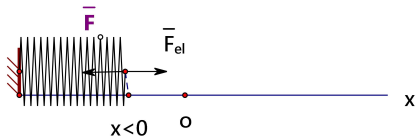
Uncompressed/
unextended



Extending



Compressing



$$\underbrace{\vec{F}_{\text{el}}}_{\text{elastic (restoring) force}} = (-kx, 0, 0)$$

elastic (restoring) force

$$\vec{F} = (kx, 0, 0)$$

$$d\vec{r} = (dx, 0, 0)$$

If the position of the right (free) end of the spring changes from x_1 to x_2



Work done by

- the external force (e.g. hand) on the spring

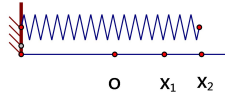
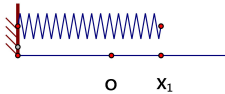
$$W_{1 \rightarrow 2}^{\text{ext}} = \int_{x_1}^{x_2} kx \, dx = \frac{1}{2} kx_2^2 - \frac{1}{2} kx_1^2$$

- the elastic force (spring) on the hand

$$W_{1 \rightarrow 2}^{\text{el}} = \int_{x_1}^{x_2} -kx \, dx = - \left(\frac{1}{2} kx_2^2 - \frac{1}{2} kx_1^2 \right) = -W_{1 \rightarrow 2}^{\text{ext}}$$

Illustration

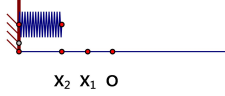
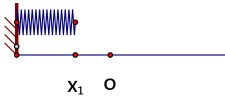
(*)



$$W_{1 \rightarrow 2}^{\text{ext}} > 0$$

$$W_{1 \rightarrow 2}^{\text{el}} < 0$$

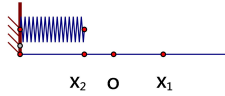
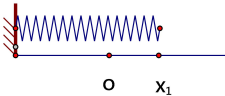
(**)



$$W_{1 \rightarrow 2}^{\text{ext}} > 0$$

$$W_{1 \rightarrow 2}^{\text{el}} < 0$$

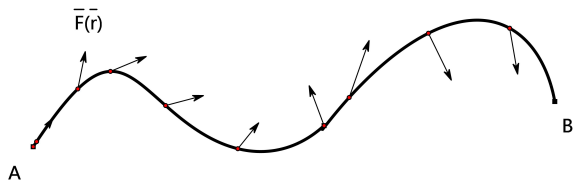
(***)



$$W_{1 \rightarrow 2}^{\text{ext}} < 0$$

$$W_{1 \rightarrow 2}^{\text{el}} > 0$$

(D) General case – varying force, curved path



$$\vec{F} = \vec{F}(\vec{r}) \neq \text{const}$$

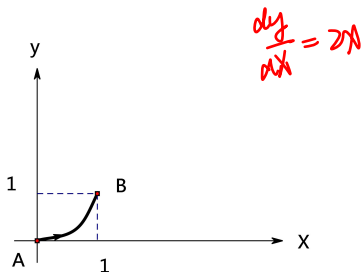
$$\delta W = \vec{F} \circ d\vec{r} \quad \text{and}$$

$$W_{AB} = \int_{\Gamma_{AB}} \vec{F} \circ d\vec{r}$$

Example (work along a parabola)

Calculate work done by the force $\vec{F}(\vec{r}) = (x^2 + y^2)\hat{n}_x + x\hat{n}_y$ [N], acting on a particle moving from (0,0) to (1,1) along a segment of parabola $y = x^2$.

Example 1. Work Along a Parabola



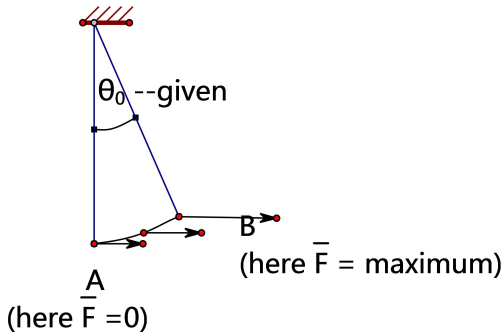
$$\Gamma_{AB} : \begin{cases} 0 \leq x \leq 1 \Rightarrow dx \\ y = x^2 \Rightarrow dy = 2x dx \end{cases}$$

Note that for all the points on the parabola $dy = 2x dx$.

$$\begin{aligned} W_{AB} &= \int_{\Gamma_{AB}} \bar{F} \circ d\bar{r} = \int_{\Gamma_{AB}} F_x dx + F_y dy = \int_{\Gamma_{AB}} (x^2 + y^2) dx + x \overset{2x \cdot dx}{\uparrow} dy \\ &= \int_0^1 [(x^2 + x^4) + (x \cdot 2x)] dx = \frac{1}{3} + \frac{1}{5} + \frac{2}{3} = \frac{18}{15} \quad [\text{J}] \end{aligned}$$

Example 2. Work Along a Curve (less abstract)

Find the minimum work needed to be done to move an object of weight Q , suspended on a light rod with length R , from A to B , acting with a horizontal force.

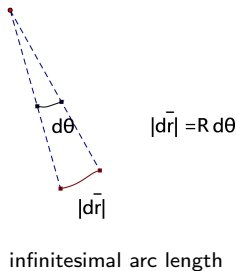
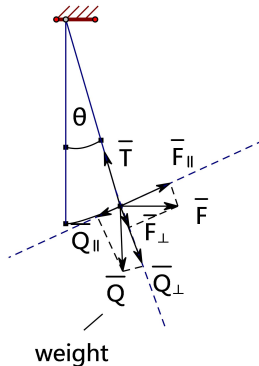


$$W_{A \rightarrow B} = \int_{\Gamma_{AB}} \vec{F} \circ d\vec{r}$$

Idea: Decompose the force

$$\delta W = \vec{F} \circ d\vec{r} = (\vec{F}_{\parallel} + \vec{F}_{\perp}) \circ d\vec{r} = \underbrace{\vec{F}_{\parallel}}_{\text{varies}} \circ d\vec{r}$$

and express everything in terms of θ



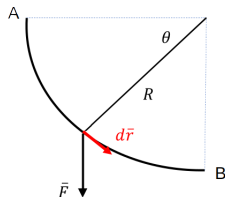
$$\delta W = F_{\parallel} |d\vec{r}| = Q \sin \theta R d\theta$$

Work done by $\vec{F}$ on the path $A \rightarrow B$ (arc of a circle with radius R)

$$\begin{aligned}\boxed{W_{A \rightarrow B}} &= \int_{\Gamma_{AB}} \vec{F} \circ d\vec{r} = \int_0^{\theta_0} \underbrace{Q \sin \theta}_{F_{\parallel}} R d\theta = QR \int_0^{\theta_0} \sin \theta d\theta \\ &= \boxed{QR [1 - \cos \theta_0]}\end{aligned}$$

Example 3. Work Done by Gravitational Force

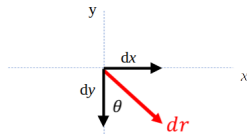
A box slides, from A to B , down a frictionless ramp in the shape of a quarter of a circle. Find work done on the box by the gravitational force.



In general

$$W_{AB} = \int_{\Gamma_{AB}} \bar{F} \circ d\bar{r}.$$

Using Cartesian coordinates, introduced as shown, we have $\bar{F} = (0, -mg)$ and $d\bar{r} = (dx, dy) = (dr \sin \theta, -dr \cos \theta)$.



Then

$$W_{AB} = \int_{\Gamma_{AB}} mg \cos \theta \, dr.$$

Just as in the previous example, for the elementary displacement along an arc of a circle, $dr = R d\theta$. So that

$$W_{AB} = \int_0^{\frac{\pi}{2}} mg \cos \theta (R d\theta)$$

And, eventually,

$$W_{AB} = mgR \int_0^{\frac{\pi}{2}} \cos \theta d\theta = mgR.$$

Work and Motion

Observations

- $W > 0 \quad \implies \quad$ particle speeds up
- $W < 0 \quad \implies \quad$ particle slows down
- $W = 0 \quad \implies \quad$ no change in speed

Conclusion. Work done by the net force implies a change in the particle's speed (or another quantity that is a function of speed).

Kinetic Energy and Work-Kinetic Energy Theorem

Recall that $\delta W = \vec{F} \circ d\vec{r}$, so the rate of work being done by the net force on a particle

$$\underbrace{\frac{\delta W}{dt}} = \vec{F} \circ \frac{d\vec{r}}{dt} = \vec{F} \circ \dot{\vec{r}} = m\vec{a} \circ \vec{v} = m\dot{\vec{v}} \circ \vec{v}$$

But $v^2 = \vec{v} \circ \vec{v}$, so $\frac{d}{dt}(v^2) = \frac{d}{dt}(\vec{v} \circ \vec{v}) = \dot{\vec{v}} \circ \vec{v} + \vec{v} \circ \dot{\vec{v}} = 2\vec{v} \circ \dot{\vec{v}}$

$$\frac{\delta W}{dt} = \frac{d}{dt} \underbrace{\left(\frac{1}{2} m v^2 \right)}_K$$

where $\boxed{K = \frac{1}{2} m v^2}$ is the **kinetic energy**. Hence,

$$\frac{\delta W}{dt} = \frac{dK}{dt} \quad \text{or} \quad \boxed{\delta W = dK}$$

Work–Kinetic Energy Theorem

Work–Kinetic Energy Theorem

Work done by the net force on a particle is equal to the change in the particle's kinetic energy.

For finite changes

$$W = \Delta K$$

$$W > 0 \quad \implies \quad \Delta K > 0 \quad \implies \quad \Delta v > 0$$

$$W < 0 \quad \implies \quad \Delta K < 0 \quad \implies \quad \Delta v < 0$$

$$W = 0 \quad \implies \quad K = \text{const} \quad \implies \quad v = \text{const}$$

- 1 Derived in a general case; no assumptions about the nature of the forces or the particle's trajectory made.
- 2 Used Newton's second law; can use only in inertial FoRs.
- 3 Kinetic energy has the units of work [$\text{J} = \text{N} \cdot \text{M} = \text{kg} \frac{\text{m}^2}{\text{s}^2}$]

Power

Power – characterizes "how fast" work is being done

$$\underbrace{\frac{\delta W}{dt}}_{\text{rate of work done}} = \vec{F} \circ \vec{v} = \underbrace{P}_{\text{instantaneous power}}$$

$$\frac{W}{\Delta t} = P_{\text{av}}$$

Units: [W] (Watt)

Alternative units: $\underbrace{\text{hp}}_{\text{horse power}} = 746W = 0.746kW$

Example (*Power of engines in vehicles*): Abrams battle tank 1500 hp, Volvo C30 108hp, Harley-Davidson 1600ccm 63 hp, Porsche 911 GT2 523 hp, Boeing 747-300 (@ 0.84 Mach, 35k feet) 330 000 hp, Diesel locomotive 5000 hp